AI risks and the importance of AI equity and balance within a team

Sadie Bartholomew

CMS Weekly Meeting
2026-10-15

National Centre for Atmospheric Science
NATURAL ENVIRONMENT RESEARCH COUNCIL

University of Reading

1. AI Risks via the Risk Register

The Institute for Research Software AI Risk Register

National Centre for Atmospheric Science
NATURAL ENVIRONMENT RESEARCH COUNCIL

(Gen)AI has pros and cons, like any technology

- Assuming we've already seen some lovely pros covered and demonstrated, let's **focus in on cons in the forms of risks, i.e. potential cons** (at least if not managed correctly / foreseen)
- Focus in on cons only **in a practical sense**, for either:
    - Our jobs: Research Software Engineering, HPC and Data Science etc., in general
    - Our field: Earth esp. Atmospheric Science as a domain, in particular
- So notably ignoring data leak/security issues (sensitive or protected information can be shared when it shouldn't be) and carbon/eco cost - though please bear the latter in mind and try to *avoid excessive use* for that reason, for one! See https://www.technologyreview.com/2025/05/20/1116327/ai-energy-usage-climate-footprint-big-tech/ for a good article considering of the environmental impact of AI use



The Institute for Research Software AI Risk Register

- “A community-maintained register of risks and supporting resources for responsible AI in Research Software Engineering” maintained by the Institute for Research Software (formerly SSI): https://jshng-glasgow.github.io/Responsible-AI-Risk-Register/
- Classifies risks by their likelihood, severity and reach; and gives topic tags, suggests mitigations and good practice, and notes ‘ownership’ i.e. who is responsible for managing the risk, all for the context of *Research* Software in particular. Submit your own via GitHub Issue if you see a gap!
- Examples of particular note for our work **[move to live view of browser]**:
  - Developing research software code that only AI understands
  - Accumulation/scaling of technical debt
  - Generated code less performant/efficient & hardware-aware
  - AI-generated [code review] hallucinating issues can waste the team's time

Technical risks

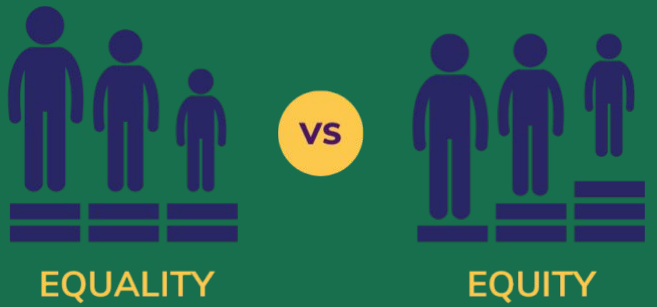
  - AI-generated code can lead to a reduced logical development and design thinking ability of the author

People/career related risks

2. AI equity* and balance within a team

Enabling everyone to progress together in the new GenAI era

*Equality means giving everyone the exact same resources, tools, or opportunities. Equity means recognizing that people start from different places and allocating specific support or resources based on individual needs to reach an equal outcome



National Centre for Atmospheric Science
NATURAL ENVIRONMENT RESEARCH COUNCIL

A plea for people to keep in mind!

It can easily take much longer to review AI-generated code (or work more generally) than to generate it. **Be mindful of your reviewer(s)!**

- GenAI can generate code very quickly. It can also majorly streamline code/workflow review, but I feel strongly that there should *always be human verification on anything* too (rather than reviewing purely by AI), and this takes time and effort
- Please consider how you can make your GenAI* work as easy to review as possible. Suggestions:
    - Break code changes down into manageable chunks per commit(/unit) with a commit message making the intention as clear as possible
    - Keep your PRs small and focused - ideally tied to a specific Issue
    - Keep a log of your and your GenAI's thoughts and share a condensed/summarised version with the reviewer. For larger-scale work, make extra notes to aid the reviewer.

*Also applies in general of course!



Analogy 1: A bike race

(With apologies to anyone who is into cycling - I’m no expert!)

Racer A Bike



Images: all from Unsplash

Analogy 1: A bike race

(With apologies to anyone who is into cycling - I’m no expert!)

Racer A Bike



Racer
B Bike



Images: all from Unsplash

Analogy 1: A bike race

(With apologies to anyone who is into cycling - I'm no expert!)

Racer A Bike



Racer C Bike



Racer
B Bike



Images: all from Unsplash

Analogy 2: more problematic bike race

Racer A



≙ No access to AI or know-how to use it effectively

Racer B



+ sneaky motor, power 250W (minor advantage)

≙ Access to only free-tier Gen AI chatbots, with minimal experience/training in its use

Racer C



+ sneaky motor, power 750W (major advantage)

≙ Access to paid-tier Gen AI tools, including advanced models and features, combined with substantial experience and training in their effective use.

Images: all from Unsplash

AI Tool Equity now!



- We should take care to ensure we all as a team have access to the same (Gen)AI tools and plans, plus time and training into their effective use (except for justifiable differences in access e.g. if someone's specific work/project means it is useful to have extra in some sense)
- The alternative is imbalance and related tension across the team going forward
- At present I think there is an imbalance and this meeting is helping to address it, thank you. But work remains to be done:
    - anyone feeling behind on GenAI use to feel supported to get up to speed
    - regular knowledge, setup and tip sharing

In summary

- (Gen)AI has pros and cons, like any technology. We briefly considered some risks i.e. potential cons, that with correct management can be averted or mitigated
- The Institute for Research Software (formerly SSI) maintains an open community resource ‘Risk Register’ which classifies and tags risks and which anyone can contribute to. See: https://jshng-glasgow.github.io/Responsible-AI-Risk-Register/, I strongly encourage you to read through this
- I wanted to highlight one risk that (Gen)AI has that I think is important to consider as a team unit: the potential to create tension from inequitable access and capability
- We should take care to ensure we all as a team have access to the same (Gen)AI tools and plans, plus time and training into their effective use
- Let’s keep sharing knowledge, setups and tips in this area to support a balance

